ASU Micro Theory Qualifier
Module 1: Consumer Theory, Demand, and Comparative Statics

Built with the PhD Qual Lecture Notes Builder workflow
Class-note integrated version

May 9, 2026

Abstract

This packet is an exam-facing module for ASU Micro Theory. It is not a textbook chapter. Its purpose is to make the standard consumer-theory and comparative-statics problems recognizable under time pressure and to give reusable proof scripts. The target weakness is definitions, mathematical intuition, and proof writing. The repeated answer architecture is

$$\text{Object} \quad \longrightarrow \quad \text{Feasibility} \quad \longrightarrow \quad \text{Optimality / Equilibrium} \quad \longrightarrow \quad \text{Verification.}$$

Depth	Exhaustive but exam-facing
Primary topics	Demand, duality, WARP/CLD, KKT, SEU, single crossing, value functions, and cost/profit analogues
Study use	Two 3-hour learning blocks plus one 90-minute exam-simulation block
Main output	Overleaf-ready TEX packet

Contents

1 Source inventory

1.1 Inventory summary

Source	Type	Topics extracted	Use in packet
Manifest	Project manifest	Required Module 1 coverage: quasilinear demand, revealed-preference addition, KKT, SEU, endogenous endowment, single crossing, value-function convexity, cost/profit monotonicity.	Determines module scope and mandatory derivations.
Builder skill	Builder workflow	Source inventory, pattern analysis, blueprint, derivation templates, exercise bank, answer sketches, quality audit, Overleaf-ready output.	Determines packet architecture.
Classes 1–2	Professor class notes	Preference primitives, budget sets, demand correspondences, homogeneity, representation, Weierstrass existence, LNS budget balance.	Definitions, baseline demand theorem, proof-writing conventions.
Classes 3–5	Professor class notes	KKT, envelope theorem, Roy's identity, indirect utility properties, expenditure minimization, Hicksian demand, Shephard lemma, concavity of expenditure.	Duality and derivative templates.
Classes 6–7	Professor class notes	Slutsky equation, budget balance, homogeneity, WARP, compensated law of demand, tax comparison, aggregate demand and Gorman form.	Revealed-preference and demand-testability section.
Classes 8–11	Professor class notes	Profit maximization, production sets, production functions, single crossing, increasing differences, Topkis, cost function properties.	Firm-side analogues and monotone comparative statics.
Classes 12–13	Professor class notes	Two-good quasilinear economy, aggregate cost minimization, aggregate utility maximization, representative consumer/-firm, excess demand in quasilinear GE.	Used lightly here; full equilibrium treatment deferred to Module 2.
Classes 14–18	Professor class notes	vN–M independence, expected utility as affine in probabilities, Allais, risk aversion, portfolio demand, Arrow–Pratt, increases in risk, SEU over acts, Ellsberg.	SEU, risk-aversion, and risky-asset subsections.
Past ASU quals 2013–2024	Exams	Quasilinear demand, Giffen/normality, single crossing, state-contingent SEU, endogenous endowment, cost/input laws, risky asset demand.	Determines exam-facing emphasis and practice map.
GitHub PE/Decision Theory folders	Course-note repository	Folder-level note inventory for demand, duality, envelopes, cost/profit, risk and SEU.	Used as source orientation; uploaded class notes supersede the earlier extraction limitation.

1.2 Class-note crosswalk

Class	Main content	Exam-facing theorem or move	Module location
1	Consumption set, budget set, preferences, demand correspondence, homogeneity of budget and demand.	Define objects before optimizing; demand is a correspondence.	Sections 4–5.1.
2	Preference representation, UM problem, existence, LNS budget balance.	Weierstrass plus LNS proof script.	Section 5.1.
3	KKT, indirect utility, envelope theorem, Roy's identity.	Use value derivatives only when it is genuinely indirect utility.	Sections 5.2 and 5.3.
4	Indirect utility properties and expenditure function.	Indirect utility is nonincreasing in prices, strictly increasing in wealth, homogeneous degree zero, quasiconvex.	Section 5.4.
5	Shephard lemma and expenditure curvature.	Expenditure is concave in prices; Hicksian Jacobian is symmetric negative semidefinite.	Section 5.4.
6	Slutsky equation and WARP.	Demand testability: budget balance, homogeneity, Slutsky matrix restrictions.	Sections 5.5 and 5.7.
7	Compensated law of demand and aggregate demand.	Add revealed-preference inequalities; CLD; Gorman aggregation as a trap-aware aside.	Section 5.5.
8	Firm problem and production plan notation.	Profit function as support function; producer supply law.	Section 5.14.
9	Scalar monotone comparative statics and SSCP.	Difference-of-differences proof, not blind IFT.	Section 5.12.
10	Vector monotone comparative statics, SID, Topkis, supermodularity.	Strong sufficient conditions for monotone choices.	Section 5.12.
11	Cost functions and returns to scale.	Cost is concave and homogeneous degree one in input prices; input-demand law.	Section 5.14.
12–13	Quasilinear economy and representative consumer/-firm.	Partial-equilibrium demand logic feeds GE/excess-demand logic.	Pointers in Sections 5.6 and 12.
14–15	vN–M expected utility, independence, Allais.	EU means affine in probabilities; WARP is not EU/SEU.	Section 5.9.
16–18	Risk aversion, Arrow–Pratt, portfolio demand, SEU acts, Ellsberg.	Concavity = risk aversion; more risk aversion lowers risky demand; SEU imposes common beliefs.	Sections 5.9–5.10.

1.3 Past-qual anchor inventory

Exam	Question	Relevant content	Core method
June 2018	Q1–Q2	Quasilinear utility $f(x_1) + x_2$, quasiconcavity iff concavity of f, closed-form demand for $-e^{-x_1} + x_2$, no-Giffen proof.	1D reduction; endpoint checks; add optimality inequalities.
August 2018	Q2	$f(x,t) = tg(x) + h(x)$, quadratic closed form, SSCP, monotone argmax, value-function curvature.	Difference of differences; supremum of affine functions.
June 2020	Q1	$u = \phi(x_1) + x_2$, $p = (1,1)$, $\sqrt{x_1}$, normality existence and counterexample to strong selection statement.	Interval expansion; quantifier discipline.
June 2024	Q1	Indirect utility $V(p, p \cdot \omega)$ and endogenous endowment choice over compact set C.	Strict wealth monotonicity; support-function inequalities.
August 2024	Q4	Two-state demand with LNS, strict quasiconcavity, WARP versus SEU-style restrictions.	KKT or revealed preference depending assumptions.
June 2022	Q2	Two-state SEU with securities and price changes.	Direct swap argument; KKT fallback with corner care.
August 2021	Q4	Quadratic vN–M utility and risky asset demand.	Mean-variance algebra; projection onto feasible interval.
June 2016	Q4	Input demand law and output monotonicity.	Revealed-cost inequalities; Topkis.
August 2015	Q2	CRS cost homogeneity.	Scale production plans both directions.
August 2013	Q3	Endowment demand and price fall for a net buyer.	Revealed preference and budget logic.

Exam-facing note

The uploaded class notes remove the earlier limitation that the GitHub PDFs were not text-extracted. This version uses the class summaries directly for notation and course framing, while still letting past quals determine which tools receive the most space.

2 Detailed blueprint

2.1 Module objective

By the end of this module you should be able to do the following under exam time pressure:

1. define the budget set, Walrasian demand, indirect utility, expenditure function, Hicksian demand, WARP, CLD, SSCP, increasing differences, SEU, risk aversion, profit, and cost;
2. reduce any two-good quasilinear demand problem to a one-dimensional global maximization problem and check corners;
3. prove comparative statics by adding optimality inequalities instead of relying on differentiability;
4. write KKT conditions with nonnegativity multipliers and use them to handle state-contingent consumption;
5. separate WARP, EU, and SEU restrictions;
6. prove monotone comparative statics using single crossing or increasing differences;
7. prove value-function convexity from the "supremum of affine functions" argument;
8. transfer the same proof moves to profit maximization, cost minimization, and input-demand monotonicity.

2.2 Prerequisites

You need only the following facts: compactness plus continuity gives existence; strict quasiconcavity gives uniqueness; a continuous function on a compact interval attains a maximum; a concave differentiable objective has sufficient KKT conditions; and a pointwise supremum of affine functions is convex.

2.3 Canonical models

Canonical model	**Question form**	**Reusable move**
Baseline Walrasian demand	Define/characterize demand, prove budget exhaustion, show existence.	Object-feasibility-optimality-verification.
Duality and demand properties	Roy, Shephard, Slutsky, CLD, WARP.	Envelope and revealed-preference inequalities.
Quasilinear two-good demand	$u = f(x_1) + x_2$, derive demand or prove no-Giffen/normality.	Budget exhaustion gives a 1D problem.
KKT with corners	Nonnegative consumption, state-contingent securities, risky assets.	Equality for positive goods, inequality for zero goods.
SEU and risk	Price-ratio implications, subjective beliefs, risk aversion, portfolio demand.	Common belief vector plus marginal utilities.

Canonical model	**Question form**	**Reusable move**
Endowment choice via indirect utility	$\max_{\omega \in C} V(p, p \cdot \omega)$.	Strict wealth monotonicity reduces to $\max p \cdot \omega$.
Single crossing and Topkis	Choices move with parameter; firm/consumer/signaling variants.	Difference-of-differences proof.
Value-function curvature	Curvature of $V(t) = \max_x \{tg(x) + h(x)\}$.	Supremum of affine functions.
Firm analogues	Profit supply law, cost input-demand law, CRS cost homogeneity.	Same inequalities with sign adjusted for min/max.

2.4 Closed-form catalog

High-yield closed forms include $\sqrt{x_1} + x_2$ demand, $-e^{-x_1} + x_2$ demand, $\log(1 + x_1) + x_2$ demand, the quadratic single-crossing problem $\max_{x \geq 0}(t-1)x - x^2/2$, two-state SEU first-order ratios, quadratic expected-utility risky-asset demand, CRS cost homogeneity, and input-demand laws.

2.5 Exercise plan

The exercise bank has 18 questions: 5 mechanical drills, 5 proof/characterization drills, 5 past-qual-style applications, and 3 trap/economist-claim questions. Every answer sketch states the setup, key equation, result, intuition, and common mistake.

3 Module map

Subtopic	Why it matters	Exam appearances	Core technique	Common trap
Budget sets and demand	Every proof starts by identifying the feasible set.	2015J Q1, 2017A Q2, 2018J Q1–Q2, 2020J Q1	Define $B(p, w)$, $d(p, w)$, budget balance.	Treating a correspondence as a function.
Indirect utility	Needed for Roy, endowment-choice problems, and comparative statics.	2024J Q1, 2017J Q1	Value function; monotonicity; homogeneity; quasiconvexity.	Applying Roy to a function not known to be indirect utility.
Expenditure/Hicksian demand	Produces Slutsky and compensated law of demand.	2017J Q1, class notes 4–7	Minimum expenditure; Shephard; concavity in prices.	Confusing Hicksian and Walrasian demand.
Quasilinear reduction	ASU repeatedly asks two-good quasilinear demand.	2018J Q1–Q2, 2020J Q1	Reduce to $\max_{x_1} f(x_1) - rx_1$.	Taking FOC without endpoint checks.
Revealed-preference addition	Fastest proof of no-Giffen, CLD, input-demand law.	2013A Q3, 2018J Q2, 2020J Q1	Write two optimality inequalities and add.	Losing the sign after adding.
KKT and corners	State-contingent and nonnegative problems often have zero demand.	2016A Q3, 2021A Q4, 2024A Q4	Complementary slackness.	Writing only interior FOCs.
SEU and risk aversion	Distinguishes SEU from generic WARP choices.	2022J Q2, 2024A Q4, 2021A Q4	Marginal utility per dollar plus common beliefs.	Forgetting WARP is weaker than SEU.
Endowment choice	Looks hard but reduces to a linear maximization.	2024J Q1	Strict wealth monotonicity of V.	Differentiating V unnecessarily.
Single crossing / Topkis	Appears in consumer, firm, signaling, screening.	2018A Q2, 2013J Q2, 2023J Q4	Difference of differences.	Claiming every maximizer under weak conditions.
Value-function curvature	Exams ask curvature without FOC calculations.	2018A Q2, class notes 8–9	Supremum of affine functions.	Saying max of affine is concave.

Subtopic	**Why it matters**	**Exam appearances**	**Core technique**	**Common trap**
Firm cost/profit analogues	Same proof architecture as demand.	2015A Q2, 2016J Q4, 2013J Q3	Revealed cost/profit inequalities.	Reversing the cost-law sign.

4 Definitions and mathematical intuition

4.1 Preference, budget, and demand primitives

Definition 4.1 (Preference relation)**.** A preference relation $\succeq$ on X is complete if for all $x, y \in X$, either $x \succeq y$ or $y \succeq x$. It is transitive if $x \succeq y$ and $y \succeq z$ imply $x \succeq z$. Write $x \succ y$ if $x \succeq y$ and not $y \succeq x$, and $x \sim y$ if both $x \succeq y$ and $y \succeq x$.

Definition 4.2 (Budget set)**.** For $X \subseteq \mathbb{R}^L_+$, $p \in \mathbb{R}^L_{++}$, and $w \geq 0$,

$$B(p, w) = \{x \in X : p \cdot x \leq w\}.$$

The budget set is homogeneous of degree zero in prices and wealth:

$$B(\alpha p, \alpha w) = B(p, w), \qquad \alpha > 0.$$

Definition 4.3 (Walrasian demand)**.** The Walrasian demand correspondence is

$$d(p, w) = \operatorname*{argmax}_{x \in B(p,w)} u(x) = \{y \in B(p, w) : u(y) \geq u(x)\ \forall x \in B(p, w)\}.$$

When the maximizer is unique, we may write $x(p, w)$.

Definition 4.4 (Local nonsatiation)**.** Preferences on $X \subseteq \mathbb{R}^L_+$ are locally nonsatiated (LNS) if for every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ with $\|y - x\| < \varepsilon$ and $y \succ x$.

Exam-facing note

LNS is not differentiability. LNS is the assumption that lets you prove budget exhaustion when prices are strictly positive. If an optimum leaves money unspent, a nearby better bundle remains affordable.

Definition 4.5 (Convexity of preferences)**.** A utility representation u is quasiconcave if every upper contour set is convex. It is strictly quasiconcave if for distinct x, y and $\lambda \in (0, 1)$,

$$u(x) \geq u(y) \quad \Rightarrow \quad u(\lambda x + (1 - \lambda)y) > u(y).$$

Quasiconcavity makes demand sets convex. Strict quasiconcavity makes demand single-valued. Neither statement replaces existence, which comes from compactness and continuity.

4.2 Indirect utility, expenditure, and Hicksian demand

Definition 4.6 (Indirect utility)**.** The indirect utility function is the value function of utility maximization:

$$V(p, w) = \max_{x \in B(p,w)} u(x).$$

Under continuous LNS preferences, it is nonincreasing in each price, strictly increasing in wealth, homogeneous of degree zero, and quasiconvex in (p, w).

Definition 4.7 (Expenditure and Hicksian demand)**.** For utility target $\bar{u}$, the expenditure function and Hicksian demand correspondence are

$$e(p, \bar{u}) = \min_{x \in X} \{p \cdot x : u(x) \geq \bar{u}\}, \qquad h(p, \bar{u}) = \operatorname*{argmin}_{x \in X} \{p \cdot x : u(x) \geq \bar{u}\}.$$

For fixed $\bar{u}$, $e(\cdot, \bar{u})$ is nondecreasing, homogeneous of degree one, and concave in prices.

What graders want

When a prompt says "define demand," do not write only an FOC. Write the budget set, the objective, the correspondence, and the domain. When a prompt says "state WARP," specify the price-wealth observations being compared and include the affordability premise.

4.3 WARP, CLD, and Slutsky terms

Definition 4.8 (WARP for a budget-balanced demand function)**.** Let $f : \mathbb{R}^{L+1}_{++} \to \mathbb{R}^L_+$ satisfy budget balance, $p \cdot f(p, w) = w$. It satisfies WARP if, for any (p, w) and (p', w'),

$$p' \cdot f(p, w) \leq w', \quad f(p, w) \neq f(p', w') \quad \Rightarrow \quad p \cdot f(p', w') > w.$$

Definition 4.9 (Compensated law of demand)**.** The compensated law of demand (CLD) says that for any price change Δp with $p + \Delta p \gg 0$, and compensation $\Delta w = \Delta p \cdot f(p, w)$ that makes the original choice affordable at new prices,

$$\Delta p \cdot \big[f(p + \Delta p, w + \Delta w) - f(p, w)\big] \leq 0.$$

In words: compensated price changes and quantities move opposite on average.

If demand is differentiable, the Slutsky substitution terms are

$$s_{ij}(p, w) = \frac{\partial d_i}{\partial p_j}(p, w) + d_j(p, w) \frac{\partial d_i}{\partial w}(p, w).$$

Under the standard differentiable theory, the Slutsky matrix $S = (s_{ij})$ is symmetric and negative semidefinite. Budget balance plus WARP implies negative semidefiniteness; symmetry generally needs stronger revealed-preference structure, such as SARP, except in two-good special cases.

4.4 Single crossing and increasing differences

Let $X, T \subseteq \mathbb{R}$ be ordered sets. A function $f : X \times T \to \mathbb{R}$ has increasing differences in (x, t) if for $x' > x$ and $t' > t$,

$$f(x', t') - f(x, t') \geq f(x', t) - f(x, t).$$

It has strict increasing differences (SID) if the inequality is strict. It has the strict single crossing property (SSCP) if, for $x' > x$ and $t' > t$,

$$f(x', t) \geq f(x, t) \quad \Rightarrow \quad f(x', t') > f(x, t').$$

Exam-facing note

Increasing differences is a difference-of-differences inequality. SSCP is a ranking implication. In exams, compute the difference of differences first; invoke a theorem only after showing the sign.

4.5 Expected utility, SEU, and risk aversion

A vN–M expected utility representation over lotteries is affine in probabilities:

$$U(p) = \sum_i u(x_i)p_i.$$

For monetary outcomes, expected-utility preferences are risk averse exactly when the vN–M utility u is concave. Jensen's inequality gives

$$u\left(\int z\, dF(z)\right) \geq \int u(z)\, dF(z).$$

A two-state subjective expected utility (SEU) representation over acts is

$$U(x_1, x_2) = \pi_1 u(x_1) + \pi_2 u(x_2), \qquad \pi_s > 0, \qquad \pi_1 + \pi_2 = 1.$$

Person 1 is more risk averse than person 0 in the Arrow–Pratt sense if $u_1 = T \circ u_0$ for a strictly increasing concave transformation T. When utilities are C^2, a local measure is

$$A_i(z) = -\frac{u_i''(z)}{u_i'(z)}.$$

4.6 Profit and cost

A production plan is $y \in \mathbb{R}^L$, with positive components as outputs and negative components as inputs. Given technology $Y \subseteq \mathbb{R}^L$, the profit function is

$$\pi(p) = \max_{y \in Y} p \cdot y.$$

As a supremum of linear functions of p, π is convex and homogeneous of degree one in prices.

For output vector q and input prices $w \gg 0$, the cost function is

$$c(q, w) = \min_{z \geq 0} \{w \cdot z : (q, -z) \in Y\}.$$

For fixed q, $c(q, \cdot)$ is nondecreasing, homogeneous of degree one, and concave in w.

5 Canonical models and derivations

5.1 Baseline Walrasian demand

Proposition 5.1 (Existence, budget balance, and uniqueness)**.** *Let $X = \mathbb{R}^L_+$, $p \in \mathbb{R}^L_{++}$, $w \geq 0$, and let u be continuous. Then $d(p, w)$ is nonempty and compact. If preferences are LNS, every $x \in d(p, w)$ satisfies $p \cdot x = w$. If u is strictly quasiconcave, $d(p, w)$ is single-valued.*

Proof. **Object.** Demand is

$$d(p, w) = \operatorname*{argmax}_{x \geq 0,\ p \cdot x \leq w} u(x).$$

Feasibility. Because $p \gg 0$, the budget set is closed and bounded: $0 \leq x_\ell \leq w/p_\ell$ for every ℓ. It is compact and nonempty.

Optimality. Continuity of u and compactness of $B(p, w)$ imply existence by Weierstrass. The argmax set is closed, hence compact.

Verification. If $p \cdot x < w$, LNS gives a nearby $y \succ x$. For y close enough, $p \cdot y \leq w$, contradicting optimality. If two distinct bundles x, y were both optimal under strict quasiconcavity, then every convex combination is feasible and strictly better than at least one optimum, a contradiction. □

Qualifier trap

The FOC is not the definition of demand. It is a condition that may be inapplicable at corners and insufficient without concavity. Start with the argmax correspondence.

5.2 Kuhn–Tucker template for utility maximization

Assume u is differentiable and concave. The consumer solves

$$\max_{x \in \mathbb{R}^L_+} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

Use

$$\mathcal{L}(x, \lambda, \mu) = u(x) + \lambda(w - p \cdot x) + \sum_{\ell=1}^{L} \mu_\ell x_\ell,$$

where $\lambda \geq 0$ is the budget multiplier and $\mu_\ell \geq 0$ is the multiplier on $x_\ell \geq 0$. The KKT system is

$$\begin{aligned}
\frac{\partial u(x)}{\partial x_\ell} - \lambda p_\ell + \mu_\ell &= 0, \qquad \ell = 1, \ldots, L, \\
\lambda(w - p \cdot x) &= 0, \\
\mu_\ell x_\ell &= 0, \qquad \ell = 1, \ldots, L, \\
p \cdot x \leq w, \qquad x \geq 0, &\qquad \lambda \geq 0, \qquad \mu \geq 0.
\end{aligned}$$

Equivalently,

$$x_\ell > 0 \Rightarrow \frac{\partial u(x)}{\partial x_\ell} = \lambda p_\ell, \qquad x_\ell = 0 \Rightarrow \frac{\partial u(x)}{\partial x_\ell} \leq \lambda p_\ell.$$

With concavity, these conditions are sufficient.

5.3 Indirect utility and Roy's identity

Suppose u, demand d, and the multiplier λ^* are differentiable near (p, w), and demand is single-valued. The envelope theorem gives

$$\frac{\partial V}{\partial w}(p, w) = \lambda^*(p, w), \qquad \frac{\partial V}{\partial p_i}(p, w) = -\lambda^*(p, w) d_i(p, w).$$

If $\partial V/\partial w > 0$, Roy's identity is

$$d_i(p, w) = -\frac{\partial V(p, w)/\partial p_i}{\partial V(p, w)/\partial w}.$$

Qualifier trap

Roy's identity only applies to an indirect utility function. In a qualifier problem, first verify homogeneity, monotonicity, and the value-function interpretation before using derivative ratios.

5.4 Expenditure, Shephard, Slutsky, and CLD

For a fixed utility target $\bar{u}$, expenditure minimization is

$$e(p, \bar{u}) = \min_x \{p \cdot x : u(x) \geq \bar{u}\}.$$

For each fixed x, $p \mapsto p \cdot x$ is linear. Since e is the pointwise minimum over feasible bundles, $e(\cdot, \bar{u})$ is concave in prices. If differentiable, Shephard's lemma gives

$$\nabla_p e(p, \bar{u}) = h(p, \bar{u}).$$

The bridge between Hicksian and Walrasian demand is

$$h(p, u(d(p, w))) = d(p, w), \qquad d(p, e(p, \bar{u})) = h(p, \bar{u})$$

when the relevant solution is unique.

Differentiating $d(p, e(p, \bar{u})) = h(p, \bar{u})$ yields the Slutsky equation

$$\frac{\partial d_i}{\partial p_j} = \frac{\partial h_i}{\partial p_j} - d_j \frac{\partial d_i}{\partial w}.$$

Equivalently,

$$s_{ij} = \frac{\partial d_i}{\partial p_j} + d_j \frac{\partial d_i}{\partial w} = \frac{\partial h_i}{\partial p_j}.$$

Because e is concave in prices, the Hicksian substitution matrix is negative semidefinite. If e is C^2, it is also symmetric.

5.5 Revealed-preference derivation of CLD

Let $x = f(p, w)$, let $p' = p + \Delta p$, and compensate wealth by $\Delta w = \Delta p \cdot x$, so x is affordable at $(p', w + \Delta w)$. Let $x' = f(p', w + \Delta w)$. WARP says that if $x' \neq x$, then the old budget cannot afford x':

$$p \cdot x' > w = p \cdot x.$$

Budget balance at the new prices gives

$$p' \cdot x' = w + \Delta w = p' \cdot x.$$

Subtract:

$$(p' - p) \cdot (x' - x) = p' \cdot x' - p' \cdot x - p \cdot x' + p \cdot x \leq 0.$$

Thus

$$\Delta p \cdot [f(p + \Delta p, w + \Delta w) - f(p, w)] \leq 0.$$

Exam-facing note

This proof is the same proof skeleton as the no-Giffen proof and the input-demand law. It survives corners and nondifferentiability because it uses only choices beating feasible alternatives.

5.6 Quasilinear two-good demand

Consider

$$u(x_1, x_2) = f(x_1) + x_2, \qquad X = \mathbb{R}^2_+, \qquad p = (p_1, p_2) \gg 0, \qquad w > 0.$$

Reusable template

Layer 1: Object.

$$d(p, w) = \underset{x_1, x_2 \geq 0}{\operatorname{argmax}} \{f(x_1) + x_2 : p_1 x_1 + p_2 x_2 \leq w\}.$$

Layer 2: Feasibility. Since utility is strictly increasing in x_2, budget balance gives

$$x_2 = \frac{w - p_1 x_1}{p_2}, \qquad 0 \leq x_1 \leq \frac{w}{p_1}.$$

Layer 3: Optimality.

$$D_1(p, w) = \underset{0 \leq x \leq w/p_1}{\operatorname{argmax}} \left\{ f(x) - \frac{p_1}{p_2} x \right\}.$$

Then

$$d(p, w) = \left\{ \left(x, \frac{w - p_1 x}{p_2} \right) : x \in D_1(p, w) \right\}.$$

Layer 4: Verification. Check $x = 0$ and $x = w/p_1$. If f is not concave, compare global values rather than relying on the FOC.

Closed form: $f(x) = \sqrt{x}$, $p = (1, 1)$

The problem is

$$\max_{0 \le x \le w} \sqrt{x} + w - x.$$

The derivative of $\sqrt{x} - x$ is $1/(2\sqrt{x}) - 1$, so the unconstrained critical point is $x = 1/4$. Strict concavity gives the global solution:

$$d(w) = \begin{cases} \{(w, 0)\}, & 0 \le w < 1/4, \\ \{(1/4, w - 1/4)\}, & w \ge 1/4. \end{cases}$$

At $w = 1/4$, both cases give $(1/4, 0)$.

Closed form: $f(x) = -e^{-x}$

The reduced problem is

$$\max_{0 \le x \le w/p_1} \left\{ -e^{-x} - \frac{p_1}{p_2} x \right\}.$$

The derivative is $e^{-x} - p_1/p_2$, and the objective is strictly concave. Define

$$a(p) = \max \left\{ 0, \log \frac{p_2}{p_1} \right\}.$$

Then

$$x_1(p, w) = \min \left\{ \frac{w}{p_1}, a(p) \right\}, \qquad x_2(p, w) = \frac{w - p_1 x_1(p, w)}{p_2}.$$

The max term is the lower-bound corner; the min term is the upper-bound wealth corner.

Quasiconcavity iff f is concave

Proposition 5.2. *For $u(x_1, x_2) = f(x_1) + x_2$ on $\mathbb{R}^2_+$, u is quasiconcave if and only if f is concave.*

Proof. If f is concave, then u is concave and hence quasiconcave. Conversely, suppose u is quasiconcave. Fix $a, b \ge 0$, $\lambda \in (0, 1)$, and choose M large enough that $(a, M - f(a))$ and $(b, M - f(b))$ are in $\mathbb{R}^2_+$. Both bundles yield utility M. Quasiconcavity implies their convex combination yields at least M:

$$f(\lambda a + (1 - \lambda) b) + \lambda(M - f(a)) + (1 - \lambda)(M - f(b)) \ge M.$$

Rearranging gives

$$f(\lambda a + (1 - \lambda) b) \ge \lambda f(a) + (1 - \lambda) f(b),$$

so f is concave. □

5.7 No-Giffen proof by adding optimality inequalities

Fix $(p_2, w) \gg 0$. Let $x^t = (x_1^t, x_2^t)$ solve the quasilinear problem at $p_1 = p_1^t$, $t = 0, 1$, and suppose

$$0 < p_1^1 < p_1^0.$$

The claim is $x_1^1 \geq x_1^0$: when good 1 becomes cheaper, demanded good 1 cannot decrease.

Proof. **Object.** We compare two solutions to two utility-maximization problems.

Feasibility. Budget balance gives $x_2 = (w - p_1 x_1)/p_2$ and the reduced problem above.

Optimality. At p_1^1, optimality of x_1^1 against x_1^0 gives

$$f(x_1^1) + \frac{w - p_1^1 x_1^1}{p_2} \geq f(x_1^0) + \frac{w - p_1^1 x_1^0}{p_2}$$

whenever x_1^0 is feasible at the lower price, which it is because the lower price expands the feasible interval. If x_1^1 is infeasible at the higher price, then $x_1^1 > w/p_1^0 \geq x_1^0$ and the claim is already proved. Otherwise, at p_1^0, optimality of x_1^0 against x_1^1 gives

$$f(x_1^0) + \frac{w - p_1^0 x_1^0}{p_2} \geq f(x_1^1) + \frac{w - p_1^0 x_1^1}{p_2}.$$

Add the inequalities and cancel:

$$(p_1^0 - p_1^1)(x_1^1 - x_1^0) \geq 0.$$

Verification. Since $p_1^0 - p_1^1 > 0$, $x_1^1 \geq x_1^0$. □

5.8 Normality: existence versus every maximizer

Let $p = (1, 1)$, $u(x_1, x_2) = \phi(x_1) + x_2$, and ϕ be continuous and strictly increasing. The reduced problem is

$$\max_{0 \leq x \leq w} \{\phi(x) - x\} + w.$$

Let $H(x) = \phi(x) - x$.

Proposition 5.3 (Weak normality as an existence statement)**.** *If $w^1 > w^0$ and $x^0 \in d(w^0)$, then there exists $x^1 \in d(w^1)$ such that $x_1^1 \geq x_1^0$.*

Proof. Let $y^0 = x_1^0 \in \operatorname{argmax}_{[0,w^0]} H$. The feasible interval expands to $[0, w^1]$. If no point in $(w^0, w^1]$ gives a value above $H(y^0)$, then y^0 remains a maximizer. If some point in $(w^0, w^1]$ gives a higher value, any new maximizer with that higher value must be greater than $w^0 \geq y^0$. In both cases there is a new maximizer weakly above the old one. □

Qualifier trap

The stronger every-selection statement is false. Let $\phi(x) = x$. Then $H(x) = 0$, so every $x \in [0, w]$ is optimal. At $w^0 = 1$, $(1, 0)$ is demanded; at $w^1 = 2$, $(0, 2)$ is demanded. Wealth rose but one selected maximizer has lower x_1.

5.9 Two-state SEU with nonnegative consumption

The consumer solves

$$\max_{x\in\mathbb{R}^2_+} \pi_1 u(x_1) + \pi_2 u(x_2) \quad \text{s.t.} \quad p_1x_1 + p_2x_2 \le w,$$

where $p \gg 0$, $\pi \gg 0$, $u' > 0$, and usually $u'' < 0$.

The KKT conditions imply that there is $\lambda > 0$ such that

$$x_s > 0 \Rightarrow \pi_s u'(x_s) = \lambda p_s, \qquad x_s = 0 \Rightarrow \pi_s u'(0) \le \lambda p_s.$$

If both states are interior, then

$$\frac{\pi_1}{\pi_2}\frac{u'(x_1)}{u'(x_2)} = \frac{p_1}{p_2}.$$

Strict concavity makes these conditions sufficient and demand unique.

Testable implication

Suppose $p_1^0/p_2^0 > p_1^1/p_2^1$ and at p^0 the consumer chooses $x_1^0 > x_2^0$. Under SEU with strictly concave u, the consumer must choose $x_1^1 > x_2^1$ at p^1, provided the boundary inequalities are handled correctly.

The proof idea is this. Since $x_1^0 > x_2^0$, strict concavity gives $u'(x_1^0) < u'(x_2^0)$. The KKT ratio therefore implies

$$\frac{\pi_1}{\pi_2} = \frac{p_1^0}{p_2^0}\frac{u'(x_2^0)}{u'(x_1^0)} > \frac{p_1^0}{p_2^0} > \frac{p_1^1}{p_2^1}$$

when both states are interior, with analogous inequalities at corners. If $x_1^1 \le x_2^1$, KKT at p^1 would imply $\pi_1/\pi_2 \le p_1^1/p_2^1$, contradiction.

Qualifier trap

WARP alone need not imply this SEU conclusion. SEU imposes one belief vector and one vN–M utility across all observations; WARP is only a pairwise revealed-preference consistency condition.

5.10 Risk aversion and risky-asset demand

For expected utility over money, risk aversion is concavity of u. The Arrow–Pratt theorem says that, for C^2 utilities with positive derivatives, one person is more risk averse than another if and only if the more risk-averse person's Arrow–Pratt measure is everywhere weakly larger, equivalently $u_1 = T \circ u_0$ for a strictly increasing concave T.

Quadratic expected utility

Let $u(z) = az - z^2/2$, initial wealth $w \in (0, a)$, risky holding $\alpha \in [0, w]$, and risky return x with mean μ_x and variance $\sigma_x^2 > 0$. Final wealth is $z = w + \alpha x$. Expected utility is

$$E[u(z)] = a(w + \alpha\mu_x) - \frac{1}{2}\left[(w + \alpha\mu_x)^2 + \alpha^2\sigma_x^2\right].$$

The derivative is

$$\frac{d}{d\alpha}E[u(z)] = \mu_x(a-w) - \alpha(\mu_x^2 + \sigma_x^2).$$

Thus

$$\alpha^*(w) = \begin{cases} 0, & \mu_x \le 0, \\ \min\left\{w, \dfrac{\mu_x(a-w)}{\mu_x^2+\sigma_x^2}\right\}, & \mu_x > 0. \end{cases}$$

The projection onto $[0, w]$ is essential.

5.11 Endogenous endowment choice via indirect utility

Consider

$$\max_{\omega \in C} V(p, p \cdot \omega),$$

where $C \subset \mathbb{R}^T_{++}$ is nonempty and compact, $p \gg 0$, and u is continuous and LNS.

Lemma 5.1 (Strict wealth monotonicity)**.** *If u is continuous and LNS and $p \gg 0$, then $V(p, w)$ is strictly increasing in w.*

Proof. Let $w' > w$, and take $x \in d(p, w)$. LNS and $p \gg 0$ imply $p \cdot x = w$. A small neighborhood of x remains inside $B(p, w')$. LNS gives a nearby $y \succ x$, and for y close enough $p \cdot y \le w'$. Hence $V(p, w') > V(p, w)$. □

Proposition 5.4 (Support-function comparative static)**.** *Let $\omega^t \in \operatorname{argmax}_{\omega \in C} V(p^t, p^t \cdot \omega)$ for $t = 0, 1$. Then*

$$(p^1 - p^0) \cdot (\omega^1 - \omega^0) \ge 0.$$

If only p_1 increases, then $\omega_1^1 \ge \omega_1^0$.

Proof. Since $V(p, w)$ is strictly increasing in w, the outer problem is equivalent to

$$\max_{\omega \in C} p \cdot \omega.$$

Optimality gives

$$p^1 \cdot \omega^1 \ge p^1 \cdot \omega^0, \qquad p^0 \cdot \omega^0 \ge p^0 \cdot \omega^1.$$

Add and rearrange to get the result. □

The simultaneous problem that chooses $(\omega, x) \in C \times X$ subject to $p \cdot x \le p \cdot \omega$ has solutions exactly on the graph of the demand correspondence evaluated at the optimal outer endowments.

5.12 Single crossing and monotone selections

Let

$$f(x, t) = tg(x) + h(x), \qquad x \in \mathbb{R}_+, \qquad t \ge 0,$$

where g is strictly increasing. For $x' > x$ and $t' > t$,

$$\big[f(x', t') - f(x, t')\big] - \big[f(x', t) - f(x, t)\big] = (t' - t)\big[g(x') - g(x)\big] > 0.$$

Thus f has strict increasing differences and hence SSCP.

Proposition 5.5 (Monotone argmax)**.** *Let $M(t) = \operatorname{argmax}_{x\in X} f(x,t)$, where $X \subseteq \mathbb{R}$. Suppose maximizers exist and f has strict increasing differences. If $t^1 > t^0$, $x^t \in M(t^t)$, then $x^1 \geq x^0$.*

Proof. Suppose $x^1 < x^0$. Optimality gives

$$f(x^1,t^1) \geq f(x^0,t^1), \qquad f(x^0,t^0) \geq f(x^1,t^0).$$

Adding yields

$$f(x^1,t^1) - f(x^0,t^1) + f(x^0,t^0) - f(x^1,t^0) \geq 0.$$

But strict increasing differences with $x^0 > x^1$ and $t^1 > t^0$ implies

$$f(x^0,t^1) - f(x^1,t^1) > f(x^0,t^0) - f(x^1,t^0),$$

which is the opposite strict inequality. Contradiction. □

For the August 2018 quadratic case,

$$f(x,t) = (t-1)x - \frac{1}{2}x^2, \qquad x \geq 0.$$

The unconstrained FOC is $x = t - 1$. With $x \geq 0$,

$$x^*(t) = \max\{0, t-1\}.$$

5.13 Value-function curvature

Let

$$V(t) = \max_{x\geq 0}\{tg(x) + h(x)\}.$$

For each fixed x, the function $t \mapsto tg(x) + h(x)$ is affine. Therefore V, as a pointwise supremum of affine functions, is convex:

$$\begin{aligned} V(\lambda t^1 + (1-\lambda)t^0) &= \sup_x\{\lambda[t^1 g(x) + h(x)] + (1-\lambda)[t^0 g(x) + h(x)]\} \\ &\leq \lambda \sup_x [t^1 g(x) + h(x)] + (1-\lambda)\sup_x [t^0 g(x) + h(x)] \\ &= \lambda V(t^1) + (1-\lambda)V(t^0). \end{aligned}$$

Qualifier trap

The maximum of concave functions is not generally concave. Here the correct reason is that the objective is affine in the parameter for every fixed choice.

5.14 Firm-side analogues

Profit supply law

If $y^t \in \operatorname{argmax}_{y\in Y} p^t \cdot y$, then

$$p^1 \cdot y^1 \geq p^1 \cdot y^0, \qquad p^0 \cdot y^0 \geq p^0 \cdot y^1.$$

Adding gives

$$(p^1 - p^0)\cdot(y^1 - y^0) \geq 0.$$

Cost homogeneity under CRS

If Y has constant returns to scale and $c(q, w)$ exists, then

$$c(\lambda q, w) = \lambda c(q, w), \qquad \lambda \geq 0.$$

For $\lambda > 0$, scale a feasible input for q up to produce λq and scale a feasible input for λq down to produce q. These two inequalities give equality.

Law of input demand

Fix output q. Let z^t solve the cost-minimization problem at input price w^t, $t = 0, 1$. Then

$$w^0 \cdot z^0 \leq w^0 \cdot z^1, \qquad w^1 \cdot z^1 \leq w^1 \cdot z^0.$$

Adding gives

$$(w^1 - w^0) \cdot (z^1 - z^0) \leq 0.$$

If only the wage/rental rate of input k increases, then $k^1 \leq k^0$.

6 Core derivation templates

Reusable template

Template 1: Quasilinear demand. State demand as an argmax over $\mathbb{R}^2_+$; use budget balance to solve for x_2; reduce to $\max_{x_1 \in [0,w/p_1]} f(x_1) - (p_1/p_2)x_1$; solve globally; return both goods; verify nonnegativity and corners.

Reusable template

Template 2: Add optimality inequalities. Write "choice 1 beats choice 0 at parameter 1" and "choice 0 beats choice 1 at parameter 0." Add. Cancel. Use the sign of the parameter change. State the quantifier: every pair, some selection, largest selection, or smallest selection.

Reusable template

Template 3: KKT with nonnegative goods. Use $\mathcal{L} = u(x) + \lambda(w - px) + \sum_\ell \mu_\ell x_\ell$. Write stationarity, primal feasibility, dual feasibility, and complementary slackness. Convert to marginal-utility-per-dollar equalities for positive goods and inequalities for zero goods. Invoke concavity for sufficiency.

Reusable template

Template 4: Duality. Define V, e, h. Use Roy only after confirming the function is indirect utility. Use Shephard only for expenditure. Use the identity $h(p, u(d(p, w))) = d(p, w)$ to derive Slutsky.

Reusable template

Template 5: Single crossing. Compute

$$[f(x', t') - f(x, t')] - [f(x', t) - f(x, t)].$$

If positive, use two optimality inequalities to rule out $x^1 < x^0$. Separate strict from weak conclusions.

Reusable template

Template 6: Supremum of affine functions. For fixed choice x, show the objective is affine in the parameter. Then use

$$\sup_x [\lambda A_x + (1 - \lambda) B_x] \leq \lambda \sup_x A_x + (1 - \lambda) \sup_x B_x.$$

The result is convexity, not concavity.

Reusable template

Template 7: Firm analogues. Profit maximization gives $(p^1 - p^0) \cdot (y^1 - y^0) \geq 0$. Cost minimization gives $(w^1 - w^0) \cdot (z^1 - z^0) \leq 0$. The algebra is the same; the sign changes because min replaces max.

7 Exam variations

1. **Quasilinear problems mutate by changing f.** With $\sqrt{x}$, $-e^{-x}$, or $\log(1+x)$, expect closed forms. With merely continuous f, expect global-maximization and correspondence arguments.
2. **No-Giffen mutates into normality.** No-Giffen fixes wealth and changes price; normality fixes prices and changes wealth. Both are reduced-problem arguments, but normality has a quantifier trap.
3. **Differentiable demand mutates into revealed preference.** If the exam asks for a testable implication but gives no differentiability, use WARP/CLD rather than Slutsky derivatives.
4. **KKT mutates into SEU restrictions.** A state-contingent problem may ask whether a choice pattern is possible. Write KKT, rank marginal utilities, and compare price ratios.
5. **Indirect utility with endogenous endowment looks harder than it is.** Use strict wealth monotonicity to reduce $\max_{\omega \in C} V(p, p \cdot \omega)$ to $\max_{\omega \in C} p \cdot \omega$.
6. **Single crossing appears outside consumer theory.** The same proof covers firm choices, information acquisition, signaling, screening, and contract quantities.
7. **Cost problems reverse the sign.** Demand/profit laws give "≥ 0"; cost minimization gives "≤ 0."

Reusable template

First 60 seconds on the exam. Does the problem say $f(x_1) + x_2$? Reduce to one dimension. Does it ask whether demand increases or decreases? Try adding optimality inequalities. Does it include $x \geq 0$ or states? Write KKT with complementary slackness. Does it include $V(p, p \cdot \omega)$? Reduce to maximizing $p \cdot \omega$. Does it say SSCP/SID? Compute the difference of differences. Does it ask for value-function curvature? Ask whether the objective is affine in the parameter for fixed choice.

8 Closed-form catalog

Model	**Closed form**	**Verification**
$u = \sqrt{x_1} + x_2$, $p = (1, 1)$	$x_1^*(w) = \min\{w, 1/4\}$, $x_2^*(w) = w - x_1^*(w)$.	Strict concavity; endpoint if $w < 1/4$.
$u = -e^{-x_1} + x_2$	$x_1^*(p, w) = \min\{w/p_1, \max\{0, \log(p_2/p_1)\}\}$.	FOC $e^{-x} = p_1/p_2$; lower and upper corners.
$u = \log(1 + x_1) + x_2$	$x_1^* = \min\{w/p_1, \max\{0, p_2/p_1 - 1\}\}$.	FOC $1/(1 + x) = p_1/p_2$.
$\max_{x \geq 0}(t - 1)x - x^2/2$	$x^*(t) = \max\{0, t - 1\}$.	Concave quadratic and boundary.
Two-state SEU interior	$\pi_1 u'(x_1)/p_1 = \pi_2 u'(x_2)/p_2$, $p \cdot x = w$.	Replace equality by inequality at zero.
Quadratic risky asset	$\alpha^* = 0$ if $\mu_x \leq 0$; otherwise $\alpha^* = \min\{w, \mu_x(a - w)/(\mu_x^2 + \sigma_x^2)\}$.	Concavity in α; projection onto $[0, w]$.
Endowment choice	$\omega^*(p) \in \operatorname{argmax}_{\omega \in C} p \cdot \omega$.	Strict wealth monotonicity of V.
Cost input law	$(w^1 - w^0) \cdot (z^1 - z^0) \leq 0$.	Add cost-minimization inequalities.
Supply law	$(p^1 - p^0) \cdot (y^1 - y^0) \geq 0$.	Add profit-maximization inequalities.

9 Common traps

Qualifier trap

Trap 1: Correspondence versus function. If strict quasiconcavity is missing, write $d(p, w)$ as a set. Multiple maximizers are not a nuisance; they are often the point.

Qualifier trap

Trap 2: Interior FOC without corner checks. Consumption sets are usually $\mathbb{R}^L_+$. Every FOC answer needs boundary inequalities or endpoint comparisons.

Qualifier trap

Trap 3: Concavity assumptions not given. ASU often withholds concavity deliberately. Revealed-preference addition works without concavity; FOC sufficiency does not.

Qualifier trap

Trap 4: Wrong monotonicity quantifier. "There exists a higher-demand maximizer" is weaker than "every higher-parameter maximizer is higher than every lower-parameter maximizer." Write the exact quantifier.

Qualifier trap

Trap 5: Roy/Shephard confusion. Roy applies to indirect utility; Shephard applies to expenditure. Do not take a derivative of an arbitrary function and call it demand.

Qualifier trap

Trap 6: WARP is not SEU. WARP is a choice-consistency axiom. SEU adds an affine probability structure, a common belief vector, and a common vN–M utility over states.

Qualifier trap

Trap 7: Value-function curvature backwards. A maximum of affine functions is convex. A minimum of affine functions is concave.

Qualifier trap

Trap 8: Cost law sign error. Cost minimization gives $(w^1 - w^0) \cdot (z^1 - z^0) \leq 0$; profit maximization gives $(p^1 - p^0) \cdot (y^1 - y^0) \geq 0$.

Qualifier trap

Trap 9: Ignoring existence. If the choice set is not compact or the objective is not continuous, do not assume a maximizer exists.

Qualifier trap

Trap 10: Profit homogeneity. Profit functions are homogeneous of degree one in prices, not degree zero. Demand is homogeneous degree zero in prices and wealth.

10 Exercise bank

10.1 Mechanical drills

Exercise 10.1 (Reduce quasilinear demand)**.** Let $u(x_1, x_2) = \log(1 + x_1) + x_2$, $p \gg 0$, and $w > 0$. Derive $d(p, w)$.

Exercise 10.2 (KKT)**.** A consumer has $U(x) = \pi_1\sqrt{x_1} + \pi_2\sqrt{x_2}$, $x \geq 0$, and budget $p \cdot x \leq w$. Write the full KKT system and solve for interior demand.

Exercise 10.3 (Difference of differences)**.** For $f(x, t) = t \log(1+x) - x^2$, $x, t \geq 0$, determine whether f has strict increasing differences.

Exercise 10.4 (Input-demand law)**.** Let z^t minimize $w^t \cdot z$ subject to producing fixed output q. Prove $(w^1 - w^0) \cdot (z^1 - z^0) \leq 0$.

Exercise 10.5 (Value-function curvature)**.** Let $V(t) = \max_{x\in[0,1]}\{tx^2 - \sin x\}$. Prove that V is convex in t.

10.2 Proof and characterization drills

Exercise 10.6 (No-Giffen)**.** For $u = f(x_1) + x_2$, with f continuous, prove that good 1 is not Giffen when (p_2, w) are fixed and p_1 falls.

Exercise 10.7 (Normality quantifiers)**.** For $u = \phi(x_1) + x_2$, $p = (1, 1)$, ϕ continuous and strictly increasing, prove the existence version of normality and give a counterexample to the every-selection version.

Exercise 10.8 (Monotone argmax)**.** Let $X \subset \mathbb{R}$ and let f have strict increasing differences. Prove that if $t^1 > t^0$, $x^t \in \operatorname{argmax}_X f(\cdot, t^t)$, then $x^1 \geq x^0$.

Exercise 10.9 (Indirect utility endowment choice)**.** Let $C \subset \mathbb{R}^L_{++}$ be compact and nonempty. Prove that any solution to $\max_{\omega\in C} V(p, p \cdot \omega)$ also solves $\max_{\omega\in C} p \cdot \omega$.

Exercise 10.10 (SEU implication)**.** Assume two-state SEU with $u' > 0, u'' < 0$. If at price p^0, $p^0_1/p^0_2 > 1$ and the consumer chooses $x^0_1 > x^0_2$, prove that $\pi_1 > \pi_2$.

10.3 Past-qual-style applications

Exercise 10.11 (June 2018 style)**.** For $u(x_1, x_2) = f(x_1) + x_2$, prove u is quasiconcave iff f is concave. Then derive demand for $f(x) = -e^{-x}$.

Exercise 10.12 (August 2018 style)**.** Solve $\max_{x\geq 0}(t - 1)x - x^2/2$. Then prove that $V(t) = \max_{x\geq 0}\{tg(x) + h(x)\}$ is convex whenever the maximum exists.

Exercise 10.13 (June 2020 style)**.** Let $u(x_1, x_2) = \sqrt{x_1} + x_2$, $p = (1, 1)$. Derive $d(w)$, prove weak normality, and explain why the stronger every-selection statement can fail.

Exercise 10.14 (June 2024 style)**.** Let $C \subset \mathbb{R}^T_{++}$ be compact. If $\omega^t \in \operatorname{argmax}_{\omega\in C} V(p^t, p^t \cdot \omega)$, prove $(p^1 - p^0) \cdot (\omega^1 - \omega^0) \geq 0$. State the implication when only p_1 rises.

Exercise 10.15 (August 2021 style)**.** Derive risky-asset demand for $u(z) = az - z^2/2$, initial wealth $w \in (0, a)$, risky return mean μ_x, and variance σ^2_x, with $\alpha \in [0, w]$.

10.4 Trap and economist-claim questions

Exercise 10.16 (FOC overreach)**.** Economist A says: "For $u = f(x_1) + x_2$, demand is characterized by $f'(x_1) = p_1/p_2$." Give two reasons this claim is incomplete.

Exercise 10.17 (WARP versus SEU)**.** Economist B says: "If a pair of state-contingent choices satisfies WARP, it must be rationalizable by SEU with risk aversion." Is this true?

Exercise 10.18 (Curvature claim)**.** Economist C says: "The maximum of concave functions is concave, so $V(t) = \max_x\{tg(x) + h(x)\}$ is concave." Diagnose the mistake and state the correct curvature result.

11 Answer sketches

Exercise 1: Budget balance gives $x_2 = (w - p_1 x_1)/p_2$. Maximize $\log(1+x) - rx$ on $[0, w/p_1]$, where $r = p_1/p_2$. The unconstrained FOC is $1/(1+x) = r$, so $x = p_2/p_1 - 1$. Project onto the interval.

Exercise 2: KKT: $\pi_s/(2\sqrt{x_s}) - \lambda p_s + \mu_s = 0$, $\mu_s x_s = 0$, $\lambda(w - px) = 0$. Interior gives $x_s = (\pi_s/(2\lambda p_s))^2$; use budget balance to solve λ.

Exercise 3: Difference of differences equals $(t' - t)[\log(1+x') - \log(1+x)] > 0$ for $x' > x$, so SID holds.

Exercise 4: Write $w^0 z^0 \le w^0 z^1$ and $w^1 z^1 \le w^1 z^0$. Add and rearrange. Fixed output is essential.

Exercise 5: For fixed x, $tx^2 - \sin x$ is affine in t. Supremum over $[0,1]$ of affine functions is convex.

Exercise 6: Use the two reduced optimality inequalities at p_1^1 and p_1^0. Add to get $(p_1^0 - p_1^1)(x_1^1 - x_1^0) \ge 0$.

Exercise 7: Let $H(x) = \phi(x) - x$. The feasible interval expands. Either keep the old maximizer or choose a new maximizer beyond the old interval. Counterexample: $\phi(x) = x$.

Exercise 8: Suppose $x^1 < x^0$. Add optimality inequalities. SID gives the opposite strict inequality. Contradiction.

Exercise 9: LNS makes $V(p, w)$ strictly increasing in wealth, so the outer problem is equivalent to maximizing $p \cdot \omega$ over C.

Exercise 10: Interior KKT gives $\pi_1/\pi_2 = (p_1^0/p_2^0)[u'(x_2^0)/u'(x_1^0)]$. Since $x_1^0 > x_2^0$, the bracket exceeds one; with $p_1^0/p_2^0 > 1$, $\pi_1 > \pi_2$. Corners only strengthen the result.

Exercise 11: Use the quasiconcavity-iff-concavity proof from Section 5.6. Demand is

$$x_1^* = \min\{w/p_1, \max\{0, \log(p_2/p_1)\}\}, \qquad x_2^* = (w - p_1 x_1^*)/p_2.$$

Exercise 12: $x^*(t) = \max\{0, t-1\}$. Convexity follows from the supremum-of-affine-functions proof.

Exercise 13: Demand is $(w, 0)$ for $w < 1/4$ and $(1/4, w - 1/4)$ for $w \ge 1/4$. Weak normality follows from interval expansion; every-selection normality fails with multiple maximizers.

Exercise 14: Since ω^t maximizes $p^t \cdot \omega$, add $p^1 \omega^1 \ge p^1 \omega^0$ and $p^0 \omega^0 \ge p^0 \omega^1$. If only p_1 rises, $\omega_1^1 \ge \omega_1^0$.

Exercise 15: Expected utility derivative is $\mu_x(a - w) - \alpha(\mu_x^2 + \sigma_x^2)$. Project the unconstrained optimum onto $[0, w]$, with zero demand if $\mu_x \le 0$.

Exercise 16: The claim ignores lower and upper corners and assumes differentiability and concavity. Correct method: solve the reduced problem globally over $[0, w/p_1]$.

Exercise 17: False. WARP is pairwise revealed-preference consistency; SEU imposes affine probability structure and common beliefs/utilities across observations.

Exercise 18: The maximum of concave functions need not be concave. Here V is convex because it is the supremum of affine functions of t.

12 Past-qual practice map

Exam	Q#	Use after studying	Target template
June 2018	Q1	Quasilinear quasiconcavity and closed-form demand.	Quasilinear protocol; concavity iff proof.
June 2018	Q2	No-Giffen proof for quasilinear preferences.	Add optimality inequalities.
August 2018	Q2	Single crossing, monotone argmax, value curvature.	Difference of differences; supremum of affine functions.
June 2020	Q1	Normality existence and counterexample.	Interval expansion; quantifier discipline.
June 2024	Q1	Endogenous endowment via indirect utility.	Strict wealth monotonicity; support-function inequality.
August 2024	Q4	WARP versus stronger utility restrictions in states.	KKT / revealed preference with state budgets.
June 2022	Q2	SEU security demands and price-ratio implication.	Swap argument; KKT fallback.
August 2021	Q4	Quadratic EU risky asset.	Mean-variance algebra; projection.
August 2022	Q2	More risk aversion and state-contingent choice.	More-concave transformation.
June 2016	Q4	Input demand and output monotonicity.	Revealed cost; Topkis.
August 2015	Q2	CRS cost homogeneity.	Scale feasibility both directions.
August 2013	Q3	Net buyer and price fall.	WARP/revealed preference.
June 2017	Q1	Expenditure nondifferentiability and normality.	Expenditure/demand correspondence.
August 2014	Q2	Nonmarket good, indirect utility, SSCP.	Value function plus single crossing.

13 Final study checklist

Before attempting old Module 1 quals, verify that you can do each item without notes.

- ☐ Define $B(p,w)$, $d(p,w)$, $V(p,w)$, $e(p,\bar{u})$, and $h(p,\bar{u})$.
- ☐ State exactly when budget balance holds.
- ☐ Prove existence of demand from compactness and continuity.
- ☐ Reduce $f(x_1)+x_2$ demand to one dimension and check endpoints.
- ☐ Derive demand for $\sqrt{x_1}+x_2$, $-e^{-x_1}+x_2$, and $\log(1+x_1)+x_2$.
- ☐ Prove no-Giffen by adding optimality inequalities.
- ☐ Prove weak normality and give the counterexample to every-selection normality.
- ☐ Write KKT conditions with nonnegativity multipliers.
- ☐ Use SEU KKT to prove a state-demand/price-ratio implication.
- ☐ State vN–M independence as affinity in probabilities.
- ☐ Define risk aversion and more risk aversion.
- ☐ Reduce $\max_{\omega\in C} V(p, p\cdot\omega)$ to $\max_{\omega\in C} p\cdot\omega$.
- ☐ Define WARP and CLD.
- ☐ Derive the Slutsky equation and identify substitution terms.
- ☐ Define SSCP and increasing differences; compute a difference of differences.
- ☐ Prove monotone argmax using two optimality inequalities.
- ☐ Prove value-function convexity as a supremum of affine functions.
- ☐ State profit and cost comparative-static laws with the correct signs.
- ☐ Identify whether a prompt asks for every maximizer, some maximizer, largest selection, or smallest selection.

14 Quality audit

Gate	**Status**
Source inventory explicit	Complete, now including uploaded class notes.
Detailed blueprint before full notes	Complete.
Module map table present	Complete.
Four-layer answer method used	Complete in canonical derivations.

Gate	**Status**
Required derivations covered	Quasilinear; adding inequalities; KKT; single crossing; supremum affine.
Class-note integration	Classes 1–18 mapped and used.
Duality from class notes included	Roy, expenditure, Shephard, Slutsky, CLD added.
Exam variations included	Complete.
Common traps included	Complete.
Exercise bank distribution	18 exercises across mechanical, proof, past-qual, and trap categories.
Answer sketches included	Complete.
Past-qual map with exact year/Q#	Complete for Module 1 anchors.
TeX readability	Uses standard packages, longtables, and tcolorbox callouts.
Remaining limitation	The compact 38-page playbook referenced by the manifest is not among the uploaded files, so exact playbook page citations are not included.